QUINN AHO

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EDUCATION

Wayne State University **Detroit, MI**

B.S. in Computer Science Engineering: *Aug 2022 – May 2026*

- GPA: 3.86
- Coursework: Computer Graphics, Graphical + Geometric Modeling, Game Programming, Intro to LLMs, Software Engineering.

Honors: **Congressional App Challenge Winner**, XR Midwest / HackDearborn Finalist, Edward + Bernard Gaffney Scholar.

WORK EXPERIENCE

Johns Hopkins Applied Physics Lab **Laurel, MD**

SWE Intern - Immersive Technologies and Software Solutions (Extended Engagement): *May 2024 – Present*

- Deployed real-world military training simulations in **Unity** for active defense programs.
- **Independently prototyped next-generation immersive trainers in Unreal Engine**, one prototype **received internal honors and secured five years of continued program funding.**
- Built a node-based authoring system enabling non-programmers to build structured mission logic directly in-editor.
- Optimized real-time rendering performance through shader batching, lightmap strategies, and custom editor-side profiling workflows.

Consistency AI (PhasePal) **Remote, USA**

Founding Unity Engineer: Phase Pal on Steam *Oct 2025 – Jan 2026*

- Helped ship PhasePal, an **AI desktop companion** with real-time LLM chat, animated 3D characters, voice synthesis, and local/cloud provider routing.
- Optimized LLM inference latency and built safety guardrails for real-time AI conversations across multiple provider backends.
- Engineered interactive character systems, cursor tracking, window perching, VRM model support, and state machines in **Unity**.

Corvid Technologies **Mooresville, NC**

VR Visualization Intern: *May 2023– Aug 2023*

- Developed **Unreal Engine VR visualization tools** and **IK-based avatar systems** using C++ and Blueprints.
- **Solo-produced an interactive cinematic demo** integrating **CAD-sourced geometry, presented directly to a defense stakeholder**.
- Improved debugging and profiling pipelines for real-time VR simulation environments.

GS Engineering **Houghton, MI**

Simulation Engineering Intern: *Nov 2023– May 2024*

- Built a custom **Unreal Engine plugin for KPI visualization in vehicle digital twin simulations**.
- Focused on system design for collisions, physics tuning, and gameplay logic for realistic scenarios.
- Created editor-friendly tools to support engineering workflows.

PROJECTS

Glint 3D – C++ Automated 3D Rendering Engine

Portfolio Project - GitHub *Feb 2025 – Present*

- **Built a C++ 3D engine from scratch,** interactive UI, scriptable CLI, JSON ops pipeline, and headless batch rendering.
- Engineered a **multi-format asset pipeline (glTF, FBX, OBJ, PLY)** with PBR shading and BVH-accelerated raytracing.
- **Dual-use design: human-driven** scene editing through the UI, **automation-driven** workflows through CLI and headless rendering.

Blueprint Health Analyzer - Unreal Engine's First-Ever Blueprint Linter

Side Hustle - Documentation Demo Video *Nov 2025 – Present*

- **Created the first dedicated Blueprint linting plugin for Unreal Engine**, identifying quality issues and enforcing best practices.
- Led a successful **20+ person beta program,** iterating on feedback to ship production-quality analysis rules and polished UX.
- Built full web documentation with interactive Blueprint examples and video walkthroughs.

AI Niagara Plugin – Natural Language VFX Control *Jan 2025 – Present*

Portfolio Project - Under Development

- Building an **Unreal Engine plugin that generates production-ready particle VFX from natural language.**
- Full editor integration: Slate UI, cloud + local AI backends, and a custom translation pipeline from LLM output to live Niagara systems.

SKILLS

Languages: C++, C#, Python, Javascript, GLSL/HLSL, Blueprint
Engines & Frameworks: Unreal Engine 4/5, Unity, Custom Engine (OpenGL, Vulkan, WebGPU)
3D & Graphics: OpenGL, DirectX, Vulkan, WebGL2, WebGPU, PBR Rendering, Raytracing, Shader Development
Pipelines & Data: CAD Integration, Blender, SolidWorks, NX, CI/CD, Git, Perforce
Specialties: 3D Visualization Platforms, VR/AR Systems, Digital Twin Simulation, Custom Tooling & Plugins, Technical Art